GAME

PATCH NOTES

2026.10.14

GENERAL

Shop Freeze NEW

- A free FREEZE SHOP button now sits next to Refresh
- Items you do not buy come back in the same slots of the next shop, with the same rarity and price
- Slots you bought stay empty, and rows that unlock after you froze the shop get new items
- Only items for sale freeze: locked rows and bought slots are left alone
- A frozen item is replaced by a new one if it can no longer be offered, for example a one-copy item you now own
- Refreshing the shop unfreezes it
- The frozen shop is saved with your run

Quitting to the main menu and pressing CONTINUE RUN while shopping now brings back the same shop

- Bought slots, refresh cost, free refreshes left and Bonus Kaart's double purchase are all kept

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Settings page NEW

- Open it from the pause window or from the main menu
- CONTROLS tab: rebind fire, special attack and auto fire for keyboard and controller, with a RESET button
- SOUND tab: music and effects volume from 0% to 100% in steps of 10%
- Leaving with unsaved changes asks if you want to save them

Pause window

- Now a proper menu with RESUME, SETTINGS and EXIT RUN
- EXIT RUN asks if you are sure, then returns to the main menu. CONTINUE RUN restarts the level you were on
- You can now pause without waiting, and also while flying to the portal

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Main menu

- The controls picture and controls text are gone, rebinding now lives in the settings page

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Shop Refresh

- The hover text now says the cost is 15% of your minerals, which is what it always cost (it said 25%)

ITEMS



Bombastic Barrier NEW (Generic rare)

- Item cost: 225$
- Your shield explodes upon breaking creating an explosion dealing 200% (+200%) damage and stuns non-boss enemies for 2 seconds.



Barrier Booster NEW (Generic common)

- Item cost: 125$
- Increases maximum shield by 20%



Barrier Backfire NEW (Generic legendary)

- Item cost: 300$
- Missiles that strike your shield have a 25% chance to reflect back to their owner. Reflected missiles deal 400% of their damage.



Blizzard Barrier (Generic legendary)

- Renamed from Calm in chaos
- Old: Deal 25% (+25%) additional damage. This bonus is lost for 3 seconds after taking damage.
- New: Whilst your shields are above 1% health, they generate a storm of ice dealing up to 400% (+400%) damage per second based on the current hitpoints of your shield.



Distributed Heat NEW (Firefighter relic)

- Item cost: 425$
- Your flamethrower now fires in ALL directions but at reduced intensity. Flamethrower deals 25% reduced damage.



Fuel Cannister (Firefighter)

- Rarity: Rare -> Legendary
- Price: 225 -> 300



Upgrade: Focus Fire (Captain relic)

- Renamed from Module: Focus Fire
- Old: Your drones no longer fire automatically. After your laserbeam hits a target, all drones immediately fire 1 shot towards the target.
- New: After your laserbeam hits a target, all drones immediately fire 1 shot towards the target.
  - Drones now keep firing on their own as well
- No longer offered to the Firefighter
- Can now be combined with Module: Accuracy and Module: Synergize

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Electro Shredding (Captain legendary)

- Renamed from Electro Shedding
- Old: Electroshred now permanently reduces enemy armor by 1 whenever it deals damage. Losing armor increases ALL damage taken.
- New: Electroshred damage is increased by up to 100% (+100%) based on the percentage of your current shield hitpoints. Also applies to the Electroshred of your drones.



Upgrade: Overload NEW (Captain rare)

- Item cost: 225$
- Drones deal an additional 15% damage.



Upgrade: Speed NEW **(Captain rare)**

- Item cost: 225$
- Drones gain 15% increased attack speed.



Upgrade: Precision NEW **(Captain legendary)**

- Item cost: 300$
- Drones gain 20% chance to critically strike, dealing double damage.



Anion Inverter (Captain relic)

- Electroshred cooldown reduction: 60% -> 70%



Module: Electric Razor (Captain relic)

- Drone Electroshred copy cooldown: 1.5 seconds -> 0.5 seconds

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Plasma Bullets (Captain and Carrier common)

- Damage over time: 300% (+300%) -> 400% (+400%)
- Duration: 1.5 seconds -> 2 seconds



Plasma Launcher (Captain rare)

- Plasma shot damage: 200% (+200%) -> 250% (+250%)



Placeholder (Generic relic)

- Maximum stationary drones: 4 -> 8
- Drone upgrades are now twice as effective on stationary drones, for both damage and attack speed
  - Old: damage was 150% of your drone damage multiplier, and attack speed upgrades did nothing
  - New: every drone damage or attack speed bonus counts double. Without drone upgrades they deal a third less damage than before
- The shop now shows their real damage per second, including your drone upgrades

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Recycler (Captain rare)

- Recycle Part drop chance per kill: 100% -> 7.5% (+7.5%)
  - It used to drop a Recycle Part on every kill by accident



Wondrous Wisdomball (Generic relic)

- Base chance for a wondrous refresh: 20% -> 30%
- Chance added after each refresh that did not trigger: 17.5% -> 20%



Guillotine (Generic legendary)

- Removed from the game



Piercing Rounds (Generic rare)

- Removed from the game

CLASSES

Drones
- Time between shots: 0.5 seconds -> 0.75 seconds

Firefighter
- Flamethrower damage ticks: every 0.28 seconds -> every 0.22 seconds
- Ignite damage: about 18% higher
- Ignite duration: 1.75 seconds -> 2.35 seconds

ENEMIES

Enemies now get stronger smoothly during a run instead of in steps after each defeated boss
- This covers movement speed, armor, knockback, detection range and the other bonuses enemies gain
- The bonus now goes up to 7 instead of 5
- Spawn cost reduction at the highest bonus: 25% -> 35%, so more enemies appear

Shuriken Mini Boss
- Movement speed bonus per difficulty step: 0.25 -> 0.4
- Now drops a mine every 2.5 seconds

Mother Ship Mini Boss
- Movement speed bonus per difficulty step: 0.15 -> 0.3
- Maximum drones at once: 4 -> 4 plus the difficulty bonus

Laserbeam Mini Boss
- Movement speed bonus per difficulty step: 0.35 -> 0.5

Defender Mini Boss
- Movement speed bonus per difficulty step: 0.35 -> 0.4

Mirage Mini Boss
- Always has at least 1 extra clone, even at the start of a run

BOSSES

Health multiplier per monster level: x1.35 -> x1.2

Red Boss

- Health: 4500 -> 4400
- Four-directional drones per cast: 2 -> 3
- Most drones alive at once: 6 -> 7
- Drone spawn cooldown: 25 seconds -> 22 seconds plus 2 seconds per drone still alive
  - It no longer speeds up below 25% health
- Shuriken spawn interval: 10 seconds -> 9 seconds

Yellow Boss

- Health: 5000 -> 4500
- Damage: 9 -> 7
- Bosses you must defeat before it can appear: 1 -> 2

CONTROLS

Controllers now work through a new system that recognises many more controller brands

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Controllers can be plugged in and out while the game runs

- You see a CONTROLLER CONNECTED or DISCONNECTED notice
- A free controller joins on any button press
- A controller that joins mid-level plays from the next level
- Every controller gets its own ship, up to 16 controllers
- A controller that drops out can reconnect to its own ship

The game pauses when nobody can play

Controller buttons for flying

- Fire: A (rebindable)
- Special attack: B (rebindable)
- Auto fire: LB (rebindable)
- Pause: Menu button
- The button that requested the main controller is gone, the main controller is now the lowest numbered one

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Keyboard

- Fire: Space, Special attack: Q, Auto fire: E (all rebindable)
- Enter no longer fires the special attack
- The keyboard now steers player 1 even when a controller is connected
- Y no longer ends the level (it was a test shortcut)

Menus now react to the keyboard the moment a key is pressed, and to the controller stick and d-pad in every menu

- Arrow keys, Enter, Space and Escape work in menus
- Input held while a screen opens is ignored until you let go
- Score card and game over now say "Press any key to continue" right away

PERFORMANCE

The game starts noticeably faster

Damage numbers are no longer drawn, because in busy levels too many of them piled up and caused slow motion

Pictures are cached with a memory limit, so the game uses less memory

Rotated pictures now use whole degrees only

Collision checks are faster

Laser beams load faster at startup

All mini bosses are now loaded at startup, so the first one in a run no longer freezes the game

BUG FIXES

Auto attack
- No longer stops responding for a while after starting a new run or pressing CONTINUE RUN

Unplugging a controller
- No longer leaves a fire button or direction stuck

Shop and stick
- The stick no longer keeps moving the cursor in the first seconds of the shop, and the relic cursor reacts at once

Shop refresh cost
- Now includes the interest from Compound Wealth

Difficulty screen
- The cards no longer overlap on smaller windows

Game over screen
- The last picture in the list can now appear

Song progress bar
- No longer starts full at the beginning of a level

Continue run
- The level progress bar and enemy spawns now follow the normal pace after pressing CONTINUE RUN



Advanced Optics
- Is no longer offered after 8 copies, which is 100% critical strike chance



Wondrous Wisdomball
- Copying your inventory no longer also copies a Legendary on top



Corrosive Oil
- Now lowers armor once per burn instead of on every burn tick



Gasoline
- Its explosion burn now follows your ignite items

Fire Shield, lingering flames and the flamethrower
- Their ignite now follows your ignite items too



Adrenaline
- Taking damage again now refreshes the health regeneration as well as the attack speed



Bonus Kaart
- Quitting and pressing CONTINUE RUN in the shop no longer gives the double purchase back

Mini bosses
- Early mini bosses no longer pile up in the list every level, which changed the odds

Royal Guard Shieldbearer
- The limit of 10 at a time now counts Shieldbearers instead of Barricades

Enemy formations
- Formations moving left now start fully off screen instead of half on screen

Boss crashes
- Striker Boss bombing runs and Twin Boss manoeuvres no longer crash the game when their boost sound is busy
- Yellow Boss reflective blocks no longer crash on a missile without a flight path

Level portal
- The portal now only appears after you picked your relic, so you can no longer skip the relic choice

Orbiting drones and missiles
- Now keep even spacing around their target

Multiplayer
- The game over check no longer fails when all ships die in the same moment

FOR DEVELOPERS

Controller library: JInput replaced by Jamepad (SDL2), native libraries and the library path option are no longer needed

New dev test switches: stressTestSpawns, enablePortalCheatKey, exportCachePerformance, disableImageCacheBudget (all off)

Performance logging with the options game.loadStatsDir and game.cacheStatsDir

Shop state now lives in one saved model (ShopContents), with freeze logic in ShopFreezer and rolling rules in ShopRoller

Controller code split into ControllerManager (seats), ControllerInputReader and MenuInput

Repo tooling added: CHANGELOG.md, CLAUDE.md, docs/planboard, docs/releases.md, .claude skills and scripts, .vscode launch configs